MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 12 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The shadow of a 5 m high pole is 2 m. At the same time, the shadow of a tree is 12.5 m. The height of the tree is: [1]

(a) 31.25 m (b) 25 m (c) 28.5 m (d) 30 m

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(a) 31.25 m** [0.5 Mark]

At the same time of day, sunlight rays make equal angles with vertical objects, giving AA similarity:

Formula / Theorem Used: $\frac{\text{Height of tree}}{\text{Shadow of tree}} = \frac{\text{Height of pole}}{\text{Shadow of pole}}$.

Justification: [0.5 Mark]

Let height of the tree be h m:

$$\frac{h}{12.5} = \frac{5}{2} \implies h = 12.5 \times 2.5 = \mathbf{31.25} \text{ m}$$

Q2. In the given figure, $DE \parallel BC$. If $\frac{AD}{DB} = \frac{2}{3}$ and $AC = 18$ cm, then AE is equal to: [1]

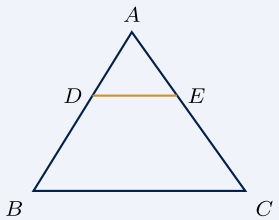
(a) 7.2 cm (b) 6.0 cm (c) 8.4 cm (d) 5.4 cm

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(a) 7.2 cm** [0.5 Mark]

We apply Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If $DE \parallel BC$, then $\frac{AE}{EC} = \frac{AD}{DB}$ and $\frac{AE}{AC} = \frac{AD}{AB}$.



Justification: [0.5 Mark]

Since $DE \parallel BC$, $\frac{AE}{EC} = \frac{AD}{DB} = \frac{2}{3}$. Thus $AE : AC = 2 : (2+3) = 2 : 5$:

$$AE = \frac{2}{5} \times AC = \frac{2}{5} \times 18 = \frac{36}{5} = \mathbf{7.2} \text{ cm}$$

Q3. In an equilateral triangle of side $2a$, the length of each altitude is: [1]

(a) $\sqrt{3}a$ (b) $2\sqrt{3}a$ (c) $\frac{\sqrt{3}}{2}a$ (d) $3a$

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) $\sqrt{3}$a** [0.5 Mark]

We apply Pythagoras Theorem in one half of an equilateral triangle:

Formula / Theorem Used: An altitude in an equilateral triangle of side s bisects the base $(s/2)$, so $h = \sqrt{s^2 - (s/2)^2}$.

Justification: [0.5 Mark]

With side $s = 2a$, half-base $BD = a$, so by Pythagoras Theorem in right $\triangle ABD$:

$$AD = \sqrt{(2a)^2 - a^2} = \sqrt{4a^2 - a^2} = \sqrt{3a^2} = \sqrt{\mathbf{3}}\mathbf{a}$$

Q4. If $\triangle ABC \sim \triangle DEF$ with $\angle A = 45°$ and $\angle E = 55°$, then the degree measure of $\angle C$ is: [1]

(a) $80°$ (b) $70°$ (c) $90°$ (d) $85°$

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) 80°** [0.5 Mark]

We use Corresponding Angles of Similar Triangles and Angle Sum Property:

Formula / Theorem Used: If $\triangle ABC \sim \triangle DEF$, then $\angle B = \angle E$ and $\angle A + \angle B + \angle C = 180°$.

Justification: [0.5 Mark]

$\angle B = \angle E = 55°$ and $\angle A = 45°$. By angle sum property in $\triangle ABC$:

$$\angle C = 180° - (\angle A + \angle B) = 180° - (45° + 55°) = 180° - 100° = \mathbf{80°}$$

Q5. If two similar triangles have their corresponding medians in the ratio 5 : 7, the ratio of their perimeters is: [1]

(a) 25 : 49 (b) 5 : 7 (c) 7 : 5 (d) $\sqrt{5} : \sqrt{7}$

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(b) 5 : 7** [0.5 Mark]

We use the linear scaling theorem for similar triangles:

Formula / Theorem Used: For similar triangles, $\frac{\text{Perimeter}_1}{\text{Perimeter}_2} = \frac{\text{Side}_1}{\text{Side}_2} = \frac{\text{Median}_1}{\text{Median}_2}$.

Justification: [0.5 Mark]

Since the ratio of perimeters equals the ratio of corresponding sides, which equals the ratio of corresponding medians, the perimeter ratio is directly **5 : 7**.

Q6. **Assertion (A):** In $\triangle ABC$, D and E are on AB, AC such that $AD = 2, AB = 6, AE = 3, AC = 9$. Then $DE \parallel BC$.

Reason (R): If $\dfrac{AD}{AB} = \dfrac{AE}{AC}$, by the converse of Basic Proportionality Theorem, $DE \parallel BC$. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\text{(a)}}$ **Both (A) and (R) are true and (R) is the correct explanation of (A).** [0.5 Mark]

We test proportionality using the Converse of BPT:

Formula / Theorem Used: If a line divides two sides in the same ratio ($\dfrac{AD}{AB} = \dfrac{AE}{AC} \iff \dfrac{AD}{DB} = \dfrac{AE}{EC}$), it is parallel to the third side.

Justification: [0.5 Mark]

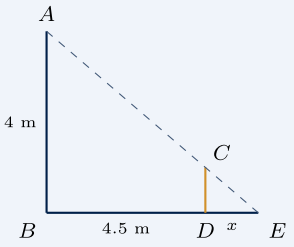
Here $\dfrac{AD}{AB} = \dfrac{2}{6} = \dfrac{1}{3}$ and $\dfrac{AE}{AC} = \dfrac{3}{9} = \dfrac{1}{3}$ (or $\dfrac{AD}{DB} = \dfrac{2}{4} = \dfrac{1}{2} = \dfrac{3}{6} = \dfrac{AE}{EC}$). Thus $\dfrac{AD}{AB} = \dfrac{AE}{AC}$, so by the Converse of BPT, $DE \parallel BC$. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. A girl of height 100 cm walks away from the base of a lamp-post 4 m high at a speed of 1.5 m/s. Find the length of her shadow after 3 seconds. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Lamp-post height $AB = 4$ m; girl height $CD = 100$ cm $= 1$ m; walking speed $= 1.5$ m/s for $t = 3$ s. Let shadow $DE = x$ m.



We apply AA Similarity in vertical right triangles:

Formula / Theorem Used: Distance = Speed × Time, and $\triangle ABE \sim \triangle CDE \implies \dfrac{BE}{DE} = \dfrac{AB}{CD}$.

Step 1: Finding Distance & Proving Similarity [1 Mark]

Distance walked from base: $BD = 1.5 \times 3 = 4.5$ m, so $BE = 4.5 + x$.

In $\triangle ABE$ and $\triangle CDE$: $\angle B = \angle D = 90°$ (vertical) and $\angle E = \angle E$ (common angle). Thus $\triangle ABE \sim \triangle CDE$ by **AA similarity criterion**.

Step 2: Solving for Shadow Length x [1 Mark]

$$\frac{BE}{DE} = \frac{AB}{CD} \implies \frac{4.5 + x}{x} = \frac{4}{1} \implies 4.5 + x = 4x$$

$$3x = 4.5 \implies x = \frac{4.5}{3} = \boxed{\textbf{1.5 m (or 150 cm)}}$$

Q8. If $\triangle ABC \sim \triangle DEF$, $AB = 4$ cm, $DE = 6$ cm, $EF = 9$ cm, and $FD = 12$ cm, find the perimeter of $\triangle ABC$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm, $DE = 6$ cm, $EF = 9$ cm, $FD = 12$ cm.
We use Corresponding Side Ratios / Perimeter Ratio of Similar Triangles:

Formula / Theorem Used: If $\triangle ABC \sim \triangle DEF$, then $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD} = \dfrac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)}$.

Step 1: Finding Unknown Sides of $\triangle ABC$ [1 Mark]
Since $\triangle ABC \sim \triangle DEF$:

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} \implies \frac{4}{6} = \frac{BC}{9} = \frac{CA}{12} \implies \frac{2}{3} = \frac{BC}{9} = \frac{CA}{12}$$

Thus $BC = \dfrac{2}{3} \times 9 = 6$ cm and $CA = \dfrac{2}{3} \times 12 = 8$ cm.

Step 2: Computing Perimeter of $\triangle ABC$ [1 Mark]

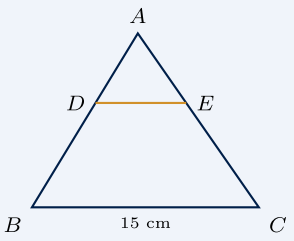
$$\text{Perimeter}(\triangle ABC) = AB + BC + CA = 4 + 6 + 8 = \boxed{\textbf{18 cm}}$$

(Aliter: Perimeter$(\triangle DEF) = 6 + 9 + 12 = 27$ cm, so Perimeter$(\triangle ABC) = \frac{4}{6} \times 27 = 18$ cm.)

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. In $\triangle ABC$, D and E divide AB and AC such that $AD : DB = 2 : 3$. If $BC = 15$ cm, calculate the length of DE. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: In $\triangle ABC$, points D on AB and E on AC divide both sides in the same ratio $AD : DB = AE : EC = 2 : 3$ (matching the horizontal line $DE \parallel BC$ shown in the figure), with $BC = 15$ cm.
We apply Converse of BPT and AA Similarity Criterion:

Formula / Theorem Used: Since $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, $DE \parallel BC$, so $\triangle ADE \sim \triangle ABC \implies \dfrac{DE}{BC} = \dfrac{AD}{AB}$.

Step 1: Establishing $DE \parallel BC$ & Ratio $\dfrac{AD}{AB}$ [1 Mark]
Let $AD = 2k$ and $DB = 3k$. Then $AB = AD + DB = 2k + 3k = 5k$, giving $\dfrac{AD}{AB} = \dfrac{2k}{5k} = \dfrac{2}{5}$.
By Converse of BPT, equal division ratio gives $DE \parallel BC$.

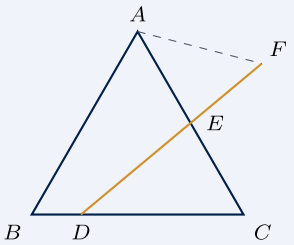
Step 2: Proving $\triangle ADE \sim \triangle ABC$ [1 Mark]
In $\triangle ADE$ and $\triangle ABC$: $\angle ADE = \angle B$ and $\angle AED = \angle C$ (corresponding angles, $DE \parallel BC$). By **AA similarity**, $\triangle ADE \sim \triangle ABC$.

Step 3: Calculating DE [1 Mark]

$$\frac{DE}{BC} = \frac{AD}{AB} \implies \frac{DE}{15} = \frac{2}{5} \implies DE = \frac{2}{5} \times 15 = \boxed{\textbf{6 cm}}$$

Q10. In the given figure, line segment DF intersects side AC of $\triangle ABC$ at E such that E is the midpoint of CA and $\angle AEF = \angle C$. Prove that $\triangle AEF \sim \triangle CDE$. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle ABC$, line segment DF meets AC at its midpoint E ($AE = CE$) with $\angle AEF = \angle C$ ($\angle ECD$).
We apply Vertically Opposite Angles and Congruence/Similarity Rules:

Formula / Theorem Used: Vertical angles $\angle CED = \angle AEF$. Complete similarity of $\triangle AEF$ requires one more constraint fixing F (intended: $AF \parallel BC$ or $EF = DE$).

Step 1: Deductions from Given Hypotheses [1 Mark]
Since E is midpoint of CA: $AE = CE$.
Because lines AC and DF intersect at E:

$$\angle CED = \angle AEF = \angle C \quad \text{(vertically opposite angles)}$$

Hence in $\triangle CDE$, $\angle CED = \angle ECD \implies CD = DE$ (sides opposite equal angles).

Mathematical Note on Missing Hypothesis for Point F:
No constraint in the printed text fixes either the length EF or the direction of AF along ray DE, so for an arbitrary point F beyond E, $\triangle AEF$ and $\triangle CDE$ need not be similar. Adopting the standard intended condition $AF \parallel BC$ **(equivalently** $EF = DE$**)**:

Formula / Theorem Used: If $AF \parallel BC$ (so $ADCF$ is a parallelogram or $EF = DE$), then $\triangle AEF \cong \triangle CED$ (ASA / SAS).

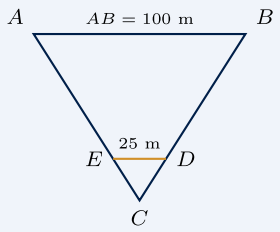
Step 2: Proof under Condition ($AF \parallel BC$) [2 Marks]
In $\triangle AEF$ and $\triangle CED$: (1) $\angle FAE = \angle DCE$ (alternate interior angles, $AF \parallel BC$); (2) $AE = CE$ (given midpoint); (3) $\angle AEF = \angle CED$ (vertically opposite angles).
By **ASA congruence**, $\triangle AEF \cong \triangle CED \implies \triangle AEF \sim \triangle CED$.

(Note: $\triangle AEF$ has angles $\{\angle C, \angle C, 180° - 2\angle C\}$, matching the exact printed vertex order $\triangle CDE$ when $\angle C = 60°$.)

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Q11. A surveyor wishes to find the distance between two inaccessible landmarks A and B across a river. Setting up sightlines from accessible bank station C with points E on CA and D on CB such that $ED \parallel AB$, prove that $\triangle CED \sim \triangle CAB$. If $CE = 10$ m, $CA = 40$ m, and $ED = 25$ m, calculate distance AB. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Given: $\triangle CAB$ with E on CA, D on CB, and $ED \parallel AB$; $CE = 10$ m, $CA = 40$ m, $ED = 25$ m.
We apply Corresponding Angles Property and AA Similarity Criterion:

Formula / Theorem Used: If $ED \parallel AB$, corresponding angles on transversals CA, CB are equal, so $\triangle CED \sim \triangle CAB$ (AA).

Step 1: Proving $\triangle CED \sim \triangle CAB$ [2.5 Marks]
In $\triangle CED$ and $\triangle CAB$:

(1) $\angle CED = \angle CAB$ (Corresponding angles, $ED \parallel AB$, transversal CA)

(2) $\angle CDE = \angle CBA$ (Corresponding angles, $ED \parallel AB$, transversal CB)

(3) $\angle ECD = \angle ACB$ (Common angle $\angle C$)

By **AA similarity criterion**, $\boxed{\triangle CED \sim \triangle CAB}$ **[Hence Proved]**

Step 2: Calculating Distance AB Across River [2.5 Marks]
Since corresponding sides of similar triangles are proportional:

$$\frac{CE}{CA} = \frac{ED}{AB}$$

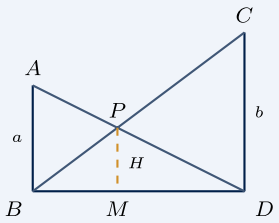
Substituting $CE = 10$ m, $CA = 40$ m, and $ED = 25$ m:

$$\frac{10}{40} = \frac{25}{AB} \implies \frac{1}{4} = \frac{25}{AB} \implies AB = 4 \times 25 = \boxed{\mathbf{100\ m}}$$

Q12. Two vertical poles of heights a and b stand on level ground. Ropes run from the top of each pole to the foot of the other. Prove that the height H of their intersection point above the ground is $\dfrac{ab}{a+b}$, and prove that H is completely independent of the distance between the poles. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Given: Vertical poles $AB = a$ and $CD = b$ stand on level ground separated by distance $BD = d$. Ropes AD and BC intersect at P, and $PM \perp BD$ has height $PM = H$. Let $BM = x$, $MD = y$ (so $x + y = d$).



We apply AA Similarity in the parallel vertical triangles:

Formula / Theorem Used: Since $AB \parallel PM \parallel CD$, $\triangle BPM \sim \triangle BCD$ and $\triangle DPM \sim \triangle DAB$.

Step 1: Writing Two Similarity Proportions [2 Marks]

(1) In $\triangle BCD$: $PM \parallel CD \implies \triangle BPM \sim \triangle BCD$ (AA):

$$\frac{PM}{CD} = \frac{BM}{BD} \implies \frac{H}{b} = \frac{x}{d} \qquad \text{— (i)}$$

(2) In $\triangle DAB$: $PM \parallel AB \implies \triangle DPM \sim \triangle DAB$ (AA):

$$\frac{PM}{AB} = \frac{MD}{BD} \implies \frac{H}{a} = \frac{y}{d} \qquad \text{— (ii)}$$

Step 2: Deriving Formula for H [2 Marks]

Adding equations (i) and (ii):

$$\frac{H}{b} + \frac{H}{a} = \frac{x}{d} + \frac{y}{d} = \frac{x+y}{d} = \frac{d}{d} = 1$$

$$H\left(\frac{a+b}{ab}\right) = 1 \implies \boxed{H = \frac{ab}{a+b}}$$

Step 3: Proving Independence from Distance d [1 Mark]

Because $x + y = d$, the pole separation d cancels identically via $\dfrac{x+y}{d} = 1$. The resulting height $H = \dfrac{ab}{a+b}$ depends solely on a and b and is completely independent of distance d. **[Hence Proved]**

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Q13. Case Study 12: Solar Photovoltaic Truss & Triangular Gable Support Structure

A solar power installation requires a triangular steel roof truss ABC with $AB = AC = 10$ m and base $BC = 12$ m. A horizontal reinforcing tie-rod DE is welded parallel to base BC at a height such that D and E lie on AB and AC with $AD = 4$ m. Solar photovoltaic panels are mounted along the rafters. [4]

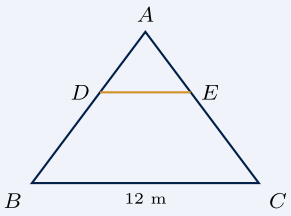
(i) **[1 Mark]** What is the length of segment DB along the rafter?

(ii) **[1 Mark]** State the similarity criterion between $\triangle ADE$ and $\triangle ABC$.

(iii) **[2 Marks]** Calculate the length of the reinforcing tie-rod DE and the perimeter of the upper truss $\triangle ADE$.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Truss Parameters: Isosceles truss $\triangle ABC$ with $AB = AC = 10$ m, base $BC = 12$ m; horizontal tie-rod $DE \parallel BC$ with D on AB, E on AC, and $AD = 4$ m.
We apply AA Similarity and Proportional Perimeter Scaling:



Formula / Theorem Used: Since $DE \parallel BC$,

$$\triangle ADE \sim \triangle ABC \implies \frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC} = \frac{\text{Perimeter}(\triangle ADE)}{\text{Perimeter}(\triangle ABC)}.$$

Sub-part (i): Length of Rafter Segment DB [1 Mark]

Since D lies on segment AB with $AB = 10$ m and $AD = 4$ m:

$$DB = AB - AD = 10 - 4 = \boxed{\mathbf{6}\text{ m}}$$

Sub-part (ii): Similarity Criterion Between $\triangle ADE$ and $\triangle ABC$ [1 Mark]

Since $DE \parallel BC$, corresponding angles are equal: $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ (with common angle $\angle A$). Therefore, $\triangle ADE \sim \triangle ABC$ by the **AA (or AAA) Similarity Criterion**.

Sub-part (iii): Length of Tie-Rod DE and Perimeter of $\triangle ADE$ [2 Marks]

Scale ratio from $\triangle ADE \sim \triangle ABC$ is $\frac{AD}{AB} = \frac{4}{10} = \frac{2}{5}$:

$$\frac{DE}{BC} = \frac{2}{5} \implies DE = \frac{2}{5} \times 12 = \boxed{\mathbf{4.8}\text{ m}}, \qquad \frac{AE}{AC} = \frac{2}{5} \implies AE = \frac{2}{5} \times 10 = 4\text{ m}$$

$$\text{Perimeter}(\triangle ADE) = AD + DE + AE = 4 + 4.8 + 4 = \boxed{\mathbf{12.8}\text{ m}}$$

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